What do you want to learn?

ALICE L WALTON SCHOOL OF MEDICINE

Bacterial Reproduction and Genetics

Last updated November 10, 2025 15 min read CC BY-SA 4.0

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Adapted from: Bacterial Genetics and Virulence Factors

Listen to this Brick
0:00 / 23:16 1x

Learning Objectives (1) Versions

After completing this brick, you will be able to:

1. Describe bacterial reproduction and genetics including mutations, transposons, and horizontal transmission mechanisms.

CASE CONNECTION

SV is calling you about a continuing urinary tract infection. "I'm not getting any better. I'm on my fourth day of the TMP-SMX medicine, and nothing has changed. I still have burning and urgency. How long is this going to take to get better?" You put her call on hold for a moment while you retrieve her lab results, including the urine culture and sensitivity.

What would explain SV's continuing symptoms? Consider your answer as you read, and we'll revisit at the end of the brick.

GO TO CONCLUSION ↓

What Are Bacterial Genetics and Virulence Factors? (LO1)

Recall that bacteria are single-celled organisms and are classified as prokaryotes. The bacteria genome is typically a single, circular chromosome made of DNA. In addition, bacteria can carry DNA in circular, extrachromosomal molecules called plasmids and in specific types of viral DNA. Bacterial DNA is haploid, meaning there is only one set of chromosome. This makes sense when we consider that bacteria undergo clonal division rather than reproduction.

Plasmids are independent of the bacterial genome and replicate independently. They most often contain genes encoding for antibiotic resistance and are extrachromosomal to facilitate ease of transfer, which we will discuss later. Bacterial cells also carry much of the same molecular machinery as human cells that are involved in replication and protein production such as ribosomes. But, because bacteria[illegible]osomes, they are common targets for antibiotics.

Bacterial and human ribosomes differ in size and structure. Bacterial ribosomes are 70S, composed of a 50S large subunit and a 30S small subunit, whereas human ribosomes are 80S, with a 60S large subunit and a 40S small subunit.

CLINICAL CORRELATION

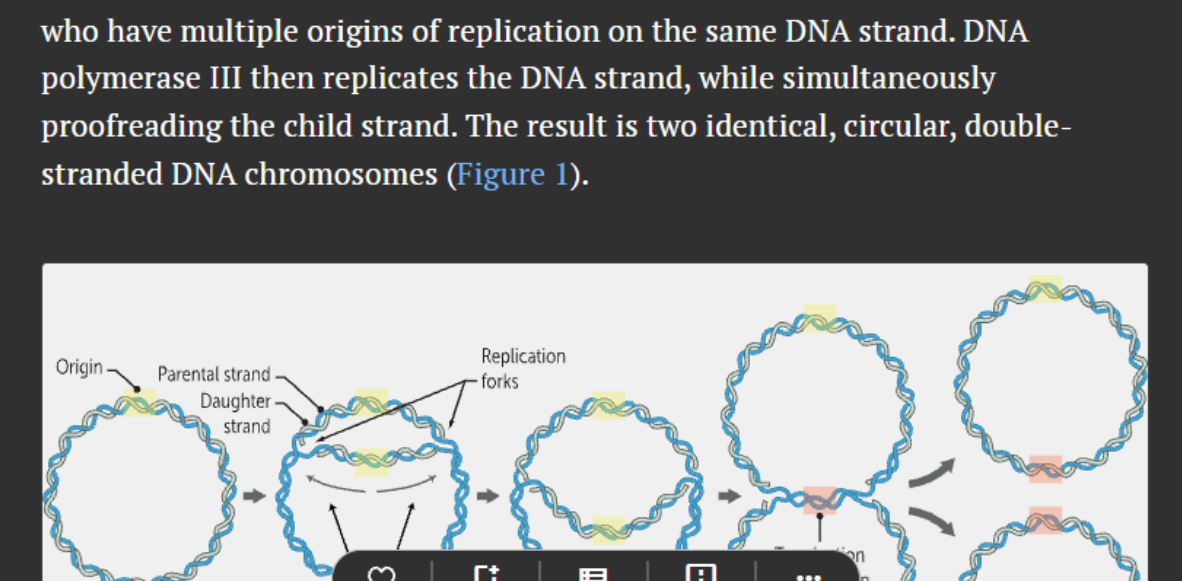
Tetracyclines, such as doxycycline, are antibiotics that target the 30S subunit of bacterial ribosomes, preventing bacterial protein production. Doxycycline is a common antibiotic used in zoonotic infections, such as Lyme disease and Rocky Mountain spotted fever.

What is the structure of bacterial DNA?

Bacterial Reproduction

Before infection can occur, bacteria have to **colonize**, or establish a colony of many bacterial cells. How do so many bacteria come about to cause an infection? Bacterial cells reproduce by binary fission, or dividing, resulting in two clones of the same bacterial cell. Thus, bacteria double their population size every cycle. That means that after 30 cycles, 1 billion bacterial cells are present. Imagine how easy this makes colonization and invasion! Think about a medieval battle: The side that has the most soldiers will usually win.

To undergo cell division, bacteria must first undergo DNA replication. Bacterial DNA replication is very similar to that in humans. However, bacterial DNA is much shorter (an approximate 5 million base pairs in *Escherichia coli* vs 245 million in humans), and, therefore, the process is simpler and quicker. The circular, chromosomal, double-stranded DNA is first unwound by a bacterial helicase, which starts at a single origin of replication, in contrast to humans who have multiple origins of replication on the same DNA strand. DNA polymerase III then replicates the DNA strand, while simultaneously proofreading the child strand. The result is two identical, circular, double-stranded DNA chromosomes (Figure 1).



QUIZ

Figure 1

In contrast to eukaryotic cells, prokaryotes have a simpler genome and may divide at a rapid rate (*E coli* is known to divide every 20 minutes!). These factors leave a lot of room for errors to happen during DNA replication. Imagine having only 20 minutes to write a paper and without autocorrect—many misspelled words would be the likely result. As we know, errors in DNA sequences give rise to spontaneous mutations. Not all mutations are bad for bacteria, because these mutations may lead to rewards such as antibiotic resistance—rewards for the bacteria, that is!

CLINICAL CORRELATION

Due to rapid replication, some bacteria can quickly overwhelm a patient before they even know something is severely wrong. In some cases of bacterial meningitis, the time between a patient's starting symptoms to death is less than 24 hrs. Even in cases where patients survive, amputations due to sepsis and permanent neurological damage is common. Vaccines are used to prevent some of these rare but extremely aggressive infections.

Bacterial Evolution

We have discussed that the combination of minimal repair mechanisms and the speed of cell division contribute to the rise of spontaneous mutations in bacterial DNA. This concept is the whole basis of the theory of evolution. In the setting of bacteria, the mutations that encode for protection against host immune responses or antimicrobial resistance should be shared and passed down. But how is this done?

Transposons are DNA sequences that contain enzyme-coding genes that allow transposition of DNA sequences within a genome (Figure 2). In bacteria, these transposons allow DNA sequences to jump around the genome as well as between plasmids and chromosomal DNA, often carrying sequences that encode for drug resistance with them. A transposon sequence must flank both sides of the gene of interest as the enzyme cleaves and ligates at the transposon sequence, leaving the gene of interest intact.

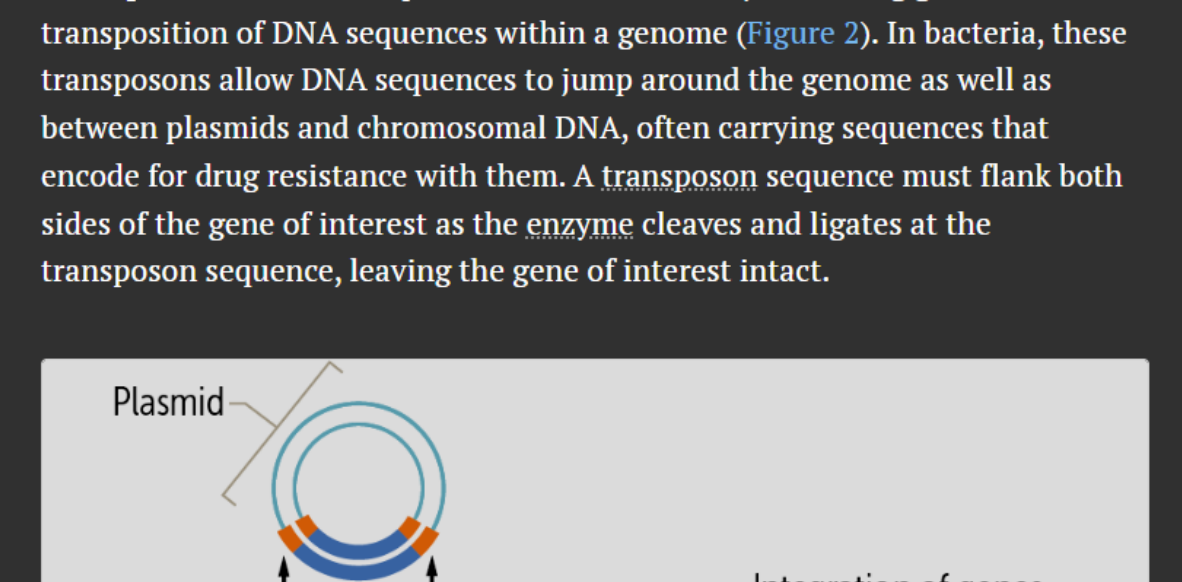


Figure 2. Transposons can carry a gene when they flank the gene.

Let's walk through a theoretical example. A person with an *E. coli* infection is being treated with ampicillin, amoxicillin, or piperacillin, which are β-lactam antibiotics used to treat gram-positive (more so than gram-negative) bacterial infections. These antibiotics function by binding to penicillin-binding proteins on bacteria and prevent cross-linking of the peptidoglycan cell wall. Under the selective pressure of antibiotic destruction, the *E. coli* bacterial cells rapidly divide. One cell may contain a gene encoding β-lactamase, which binds to penicillin molecules, degrades them, and provides bacteria with resistance to these antibiotics.

The sequence encoding β-lactamase can then be transposed from the bacterial chromosomal DNA to the plasmid if there are transposons on each side of the gene. This plasmid can then be transferred to other bacteria not containing the β-lactamase gene. This type of genetic transmission is known as conjugation, which will be discussed in detail later. This gene is then either kept as a plasmid or transferred from the plasmid to the chromosomal DNA of the receiving bacteria. From then on, all future progeny of that bacterium will have the β-lactamase gene.

CLINICAL CORRELATION

In the hospital setting, empirical antibiotic therapy may be used to treat patients with suspected infections. This is done in order to prevent the infection from progressing to sepsis, which has dire consequences and can lead to systemic disturbances such a[illegible]e commonly placed on a broad-spectrum antibiotic such as piperacillin/tazobactam to fight the infection while the work-up continues. Some argue that the commonplace use of broad-spectrum antibiotics can lead to the formation of superbugs with antibiotic resistance that are selected out through the similar concept of natural selection, when repeatedly exposed to antibiotics.

Genetic Exchange

We have discussed the ability of bacteria to grow and mutate rapidly, allowing them to readily adapt to challenges and environments. From a clinical perspective, we view this characteristic of bacteria as a means to obtain antibiotic resistance.

Recall that bacteria are simple cells that replicate and divide when they reproduce. When humans reproduce on the other hand, mixing of DNA from two different sources promotes different characteristics, or phenotypes, to appear. That's the reason no two people on earth are genetically the same, except identical twins, triplets, and so on.

Well, bacteria exchange DNA, too. Wait—so, bacteria can mate? In a manner of speaking, yes. In fact, this ability of bacteria to exchange useful mutations via genetic transfer to other bacteria allows an adaptation (such as antibiotic resistance) to spread among a population of bacteria in the human body.

We will discuss three mechanisms that allow bacteria to obtain new fragments of DNA from other bacteria. Collectively, these mechanisms are known as **horizontal gene transfer (HGT)**. Incredibly, not only can HGT mechanisms occur within the same species of bacteria, but also some bacteria are even capable of obtaining [illegible]nera.

The most common HGT mechanism is conjugation, and this mechanism plays out most closely to what you might be imagining when you think about bacteria "mating." The other two mechanisms occur less commonly, with transformation occurring more often than transduction.

Through **conjugation, transformation, and transduction**, bacteria are able to evolve. This evolution, much like our own, has allowed the survival of the fittest bacteria—that is, those that have acquired antibiotic resistance, as well as immune system resistance, making them more dangerous infectious agents.

What mechanism allows bacteria to evolve?

Conjugation. Just as the root of this word (conjoin) suggests, conjugation is how bacteria have sex, or join together. This is the process that we alluded to during the discussion of transferring plasmids with antibiotic resistance from one bacterium to ano[illegible] **plasmid** are able to undergo conjugation. The F plasmid contains DNA that encodes for the production of a sex pilus. The sex pilus on an F+ bacterium connects to another bacterium of similar or same species that does not have the F plasmid. The bacterium without the plasmid is known as F–, and, when the sex pilus of the F+ bacterium connects with the F– bacterium, it forms a mating bridge. A single strand of the double-stranded F plasmid is then transferred across the mating bridge to the other bacterium, starting at the origin of the plasmid (Figure 3). There is no transfer of chromosomal DNA. Following plasmid transfer, each bacterium synthesizes the necessary complementary strand, thus restoring the double-stranded plasmid. Plasmids usually contain genes that encode for antibiotic resistance.

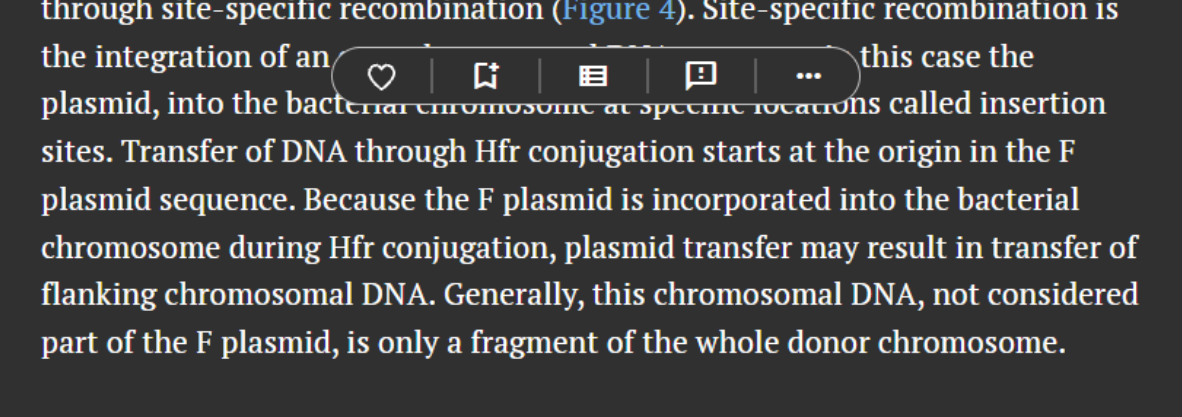


Figure 3: F+ x F- conjugation steps

Bacterial strains that are considered a high-frequency recombination strain, or Hfr, contain the F plasmid in the chromosomal DNA, usually integrated through site-specific recombination (Figure 4). Site-specific recombination is the integration of an[illegible]this case the plasmid, into the bacterial chromosome at specific locations called insertion sites. Transfer of DNA through Hfr conjugation starts at the origin in the F plasmid sequence. Because the F plasmid is incorporated into the bacterial chromosome during Hfr conjugation, plasmid transfer may result in transfer of flanking chromosomal DNA. Generally, this chromosomal DNA, not considered part of the F plasmid, is only a fragment of the whole donor chromosome.

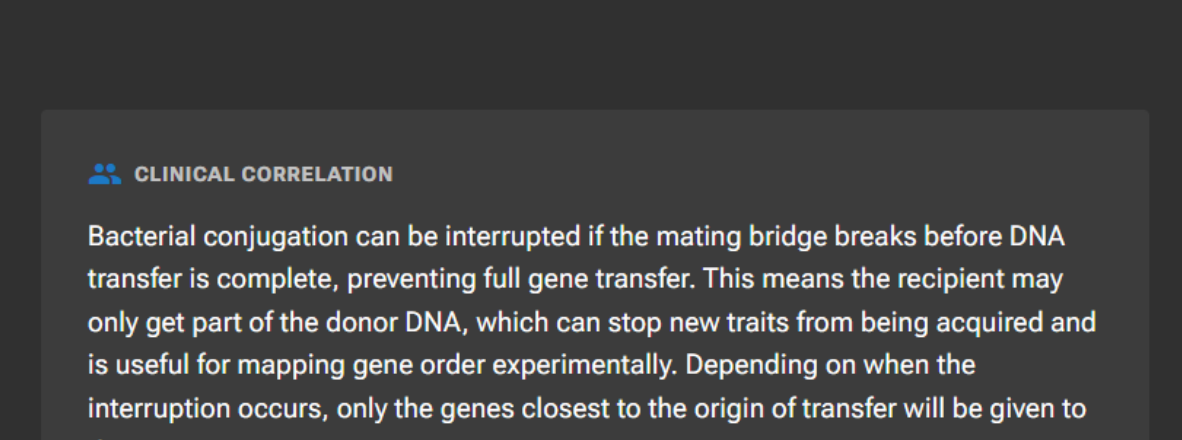

Plasmid incorporates into bacterial DNA
Transfer and replication of part of the chromosome
Hfr cell
F⁻cell
Hfr cell
F⁻cell
Hfr cell
Recombinant F⁻cell

Figure 4: Hfr (high-frequency recombination) x F- conjugation steps

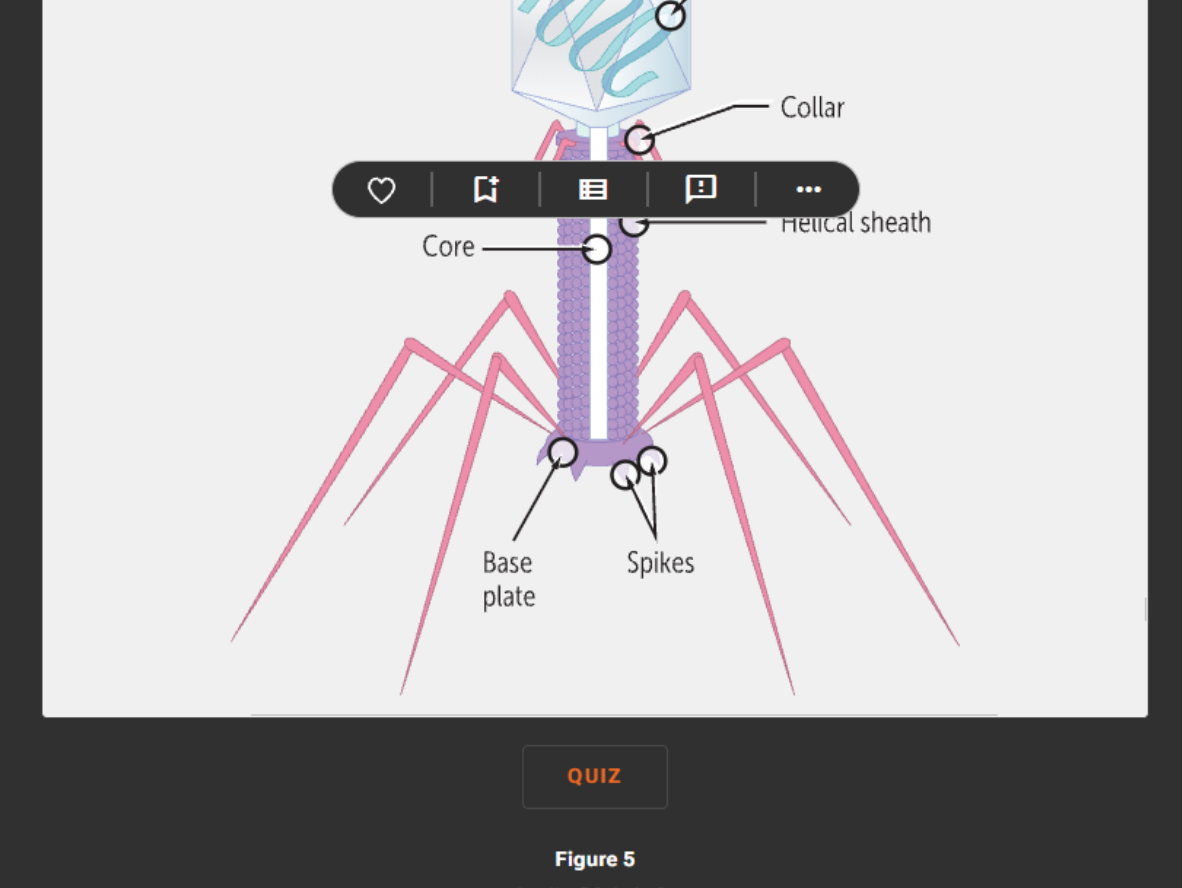
Credit: ©ScholarRx

CLINICAL CORRELATION

Bacterial conjugation can be interrupted if the mating bridge breaks before DNA transfer is complete, preventing full gene transfer. This means the recipient may only get part of the donor DNA, which can stop new traits from being acquired and is useful for mapping gene order experimentally. Depending on when the interruption occurs, only the genes closest to the origin of transfer will be given to the recipient.

Transformation. Transformation is the taking up of naked DNA from the environment. Bacteria capable of transformation are called "competent." The naked DNA usually comes from dying or lysed cells. Implementation of the absorbed bacteria into the host bacterial chromosome occurs through homologous recombination, which requires the host DNA and absorbed DNA to have similar nucleotide sequences that are then exchanged. Examples of competent bacteria are *Streptococcus pneumoniae, Haemophilus influenzae* type B, *Neisseria meningitidis*, and *E. coli*.

Transduction. Paradoxically, a bacterium, which is an invader itself, can become invaded by viruses. Viruses that infect bacteria are called bacteriophages, or just phages (Figure 5). The bacteriophage infects a susceptible bacterial cell by binding on surface receptors and injecting its DNA into the host bacterial cell. The rest of the phage - the protein coat, or capsid - remains on the outside of the bacteria. Most phages contain DNA that encodes for production of phage progeny, which ultimately results in lysis of the infected cell.



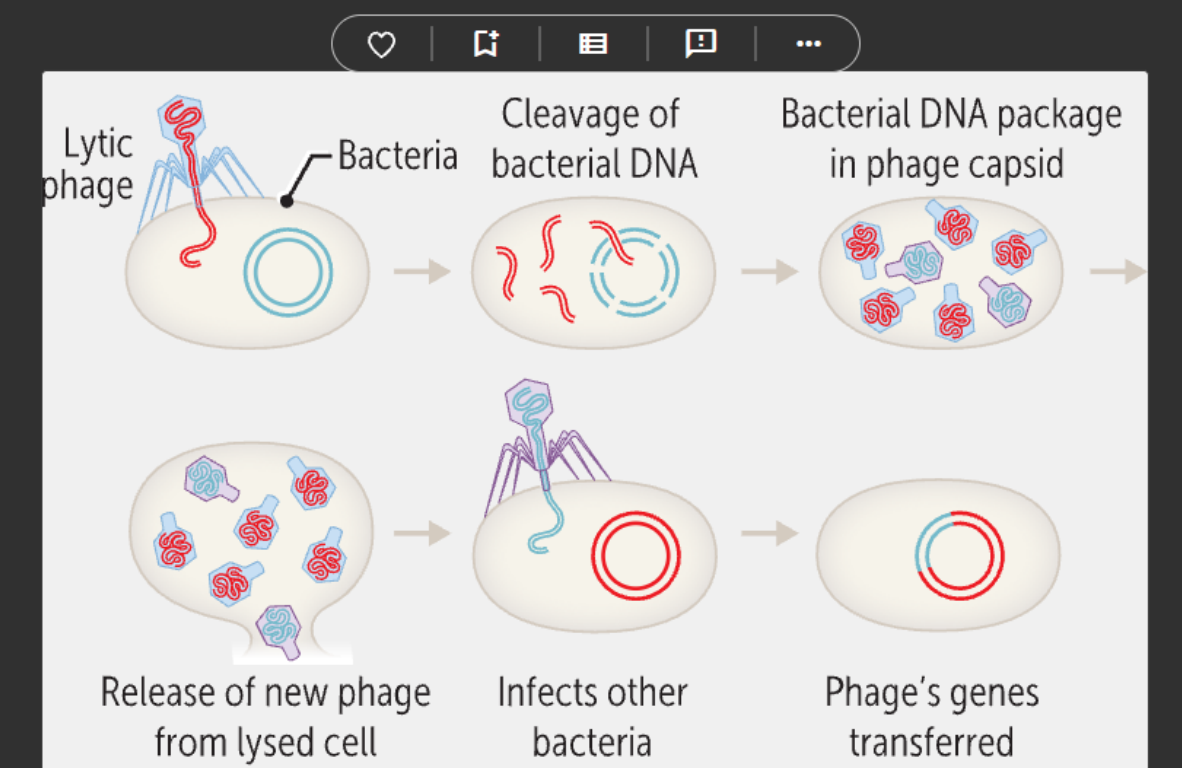
QUIZ

Figure 5

Credit: ©ScholarRx

Transduction can be thought of in two different ways: generalized transduction and specialized transduction. Generalized transduction is a packaging process, while specialized transduction is an excision process.

In generalized transduction, during lytic growth, some phages are packaged with host bacterial DNA rather than viral DNA (Figure 6). These are called transducing phages and are the phages that participate in transduction. A bacterial cell infected by a transducing phage does not result in lysis of the cell but, instead, expression of the donor bacterial DNA. Generalized transduction is swapping of genes, through homologous recombination, into the chromosome of recipient progeny.


Lytic phage
Bacteria
Cleavage of bacterial DNA
Bacterial DNA package in phage capsid
Release of new phage from lysed cell
Infects other bacteria
Phage's genes transferred

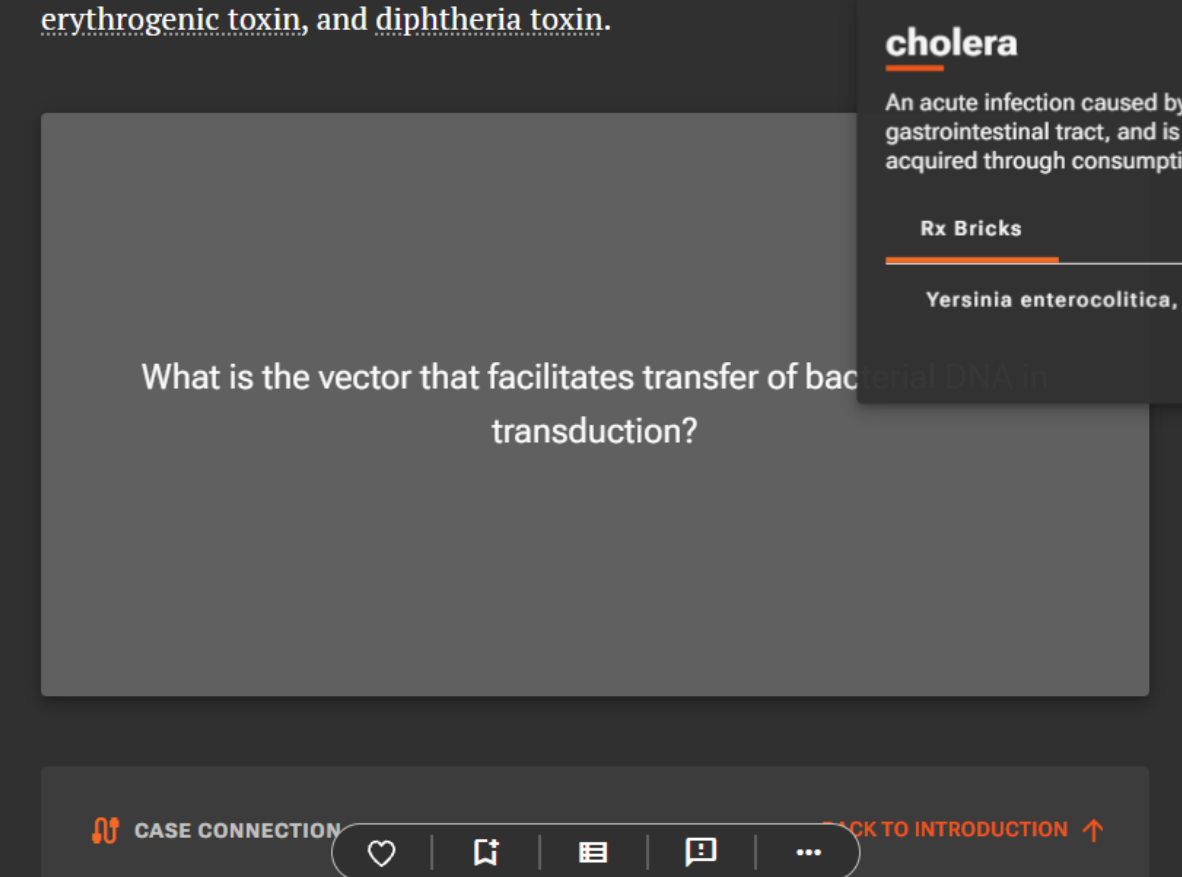
QUIZ

Figure 6: Generalized transduction. A lytic phage infects a bacterium, leading to cleaved bacterial DNA, which is packaged in the phage capsid. This new phage infects a new bacterium, transferring DNA from the donor bacterium.

Credit: ©ScholarRx

Specialized transduction is an excisional process. A temperate bacteriophage infects a bacterium and transfers viral DNA, which incorporates into the host DNA, forming a prophage (Figure 7). The viral DNA becomes dormant (doesn't actively replicate or cause lysis of host bacterial DNA) and is passed on to child cells.

When induction occurs (lysogenic cell is exposed to stimulus), the prophage DNA is imprecisely excised, bringing along the host bacterium's immediately adjacent DNA (excisional process). When bacteriophages formed from this process inject a new recipient with the hybrid DNA, that recipient bacterium's genome will then include its own DNA, bacteriophage viral DNA, and DNA from the previous bacterial host. It is important to note that these are specific portions of bacterial DNA that were adjacent to the incorporated viral DNA, as opposed to the random DNA that may be incorporated in generalized transduction.


Lysogenic phage
Viral DNA
Bacteria
Viral DNA incorporates in bacterial DNA
Phage particles carry bacterial DNA
Release of new phage from lysed cell
Infects other bacteria
Genes different from donor and recipient

QUIZ

Figure 7: Specialized transduction. Lysogenic phage infects a bacterium, and viral DNA is incorporated into the genome. When the phage genome is excised to be packaged into capsid(s), flanking bacterial DNA may be carried with it.

Temperate bacteriophages often contain DNA that encodes for proteins that are essential for bacterial virulence. The prime example is the production of bacterial toxins such as Shiga-like toxin, botulinum toxin, cholera toxin, erythrogenic toxin, and diphtheria toxin.

cholera

An acute infection caused by exotoxins from the *Vibrio cholerae* bacteria involving the gastrointestinal tract, and is marked by profuse, watery diarrhea; transmitted fecal-orally and acquired through consumption of contaminated water.

Rx Bricks

Yersinia enterocolitica, Campylobacter jejuni, Vibrio cholerae, Helicobacter pylori

Search Term

What is the vector that facilitates transfer of bacterial DNA in transduction?

CASE CONNECTION

BACK TO INTRODUCTION ↑

Thinking back to SV, what would explain her continuing symptoms?

The urine culture was positive for >100,000 CFU of *E coli*. Usua[illegible] sulfamethoxazole would treat this, but the sensitivity results sh[illegible] resistant to TMP-SMX and sensitive to various fluoroquinolone[illegible] phone. "I have the complete urine culture test results; they exp[illegible] improving. The bacteria you have is usually susceptible to the a[illegible] taking, but in your case, it's resistant to it. We can change you t[illegible] antibiotic, one shown to work against this infection." You ask S[illegible] you in 36-48 hours because she should start to see improveme[illegible] is taking this different antibiotic.

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Summary

- Many bacteria have circular, double-stranded, chromosomal DNA and circular, extrachromosomal DNA called plasmids. Some have linear chromosomes.
- Bacterial DNA replication functions similar to that in humans and is mainly mediated by DNA polymerase III.
- Rapid cell division may lead to spontaneous DNA mutations that can convey a selective advantage or disadvantage. Selective pressure allows the mutation to be selected for or against.
- Antibiotic resistance is shared purposefully through plasmids, which can then be transferred by transposons into the chromosomal DNA of the recipient.
- Conjugation transfers plasmids from one bacterium to another through physical contact.
- Transformation i[illegible]petent bacterium.
- Transduction is the transfer of DNA via the action of b[illegible]

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Review Questions

1. Which statement is true about bacterial DNA replicatio[illegible]

A. ○ It does not involve helicase enzymes.
B. ● It has only one origin of replication.
C. ○ It is slower than human DNA replication.
D. ○ It occurs within the cell's nucleus.
E. ○ The product is two linear DNA strands.

Hide Explanation

The correct answer is it has only one origin of replication (B). The process of DNA replication in bacteria is very similar to that in humans. It does not involve helicase enzymes (A) is incorrect, as it does use those enzymes to unwind the strands. DNA polymerase III then synthesizes new strands of DNA while simultaneously proofreading. Bacterial DNA replication is much faster than DNA replication in humans, making it is slower than human DNA replication (C) incorrect, as the bacterial DNA usually contains 5 million base pairs compared to approximately 245 million in humans. *E coli* cells, for example, are known to divide within minutes. Bacteria are prokaryot[illegible] making it occurs within the cell's nucle[illegible]NA is circular, double-stranded DNA, and replication results in two circular co[illegible] is two linear DNA child strands (E) is incorrect.

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2. What is the process of transferring antibiotic resistanc[illegible] contact?

A. ● Conjugation
B. ○ Replication
C. ○ Transcription
D. ○ Transduction
E. ○ Transformation

Hide Explanation

The correct answer is conjugation (A). Conjugation is the process of bacteria forming a mating bridge with another bacterium and transferring a plasmid. Replication (B) is the process of DNA replication prior to bacterial cells dividing. Transcription (C) is the process of producing RNA from DNA. Transduction (D) is the process of transferring DNA using a bacteriophage that infects bacterial cells. Transformation (E) is the process of a bacterial cell absorbing naked DNA, often from lysed cells, and incorporating it into its DNA.

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3. Which form of horizontal gene transfer results in the ev[illegible] recipient bacteria?

A. ○ Conjugation
B. ○ Transformation
C. ○ Generalized Transduction
D. ● Specialized Transduction
E. ○ Transcription

Hide Explanation

D) With specialized transduction, a piece of the host bacteria's DNA is taken when the lysogenic phage cuts its DNA from the bacteria. The next bacteria infected with this phage will get the previous host's bacteria DNA when the phage inserts into its genome. Eventually, the phage will lyse the bacteria and start the cycle over again. C) With generalized transduction, a piece of the host bacteria's DNA is accidentally packaged in a phage coat but doesn't contain the phage's DNA. The released phage coat with bacterial DNA inside is not lytic because it doesn't contain viral DNA, but it can still bind and release the previous bacteria's DNA into a new bacterium. Neither A), B), or E) results in the lysis of bacteria.

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How would you rate this Brick?
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Review Learning Objectives

Check off the objectives when you feel comfortable with the concept

01 ☑ ~~Describe bacterial reproduction and genetics including mutations, transposons, and horizontal transmission mechanisms.~~

Objective Confidence: 1/1